

MATERIALS FOR MEMORY AND DISPLAY SYSTEMS

Memory Devices: Introduction, Basic concepts of electronic memory, History of organic/polymer electronic memory devices, Classification of electronic memory devices, types of organic memory devices (organic molecules, polymeric materials, organic-inorganic hybrid materials).

Display Systems: Photoactive and electroactive materials, Nanomaterials and organic materials used in optoelectronic devices. Liquid crystals (LC's) - Introduction, classification, properties and application in Liquid Crystal Displays (LCD's). Properties and application of Organic Light Emitting Diodes (OLED's) and Quantum Light Emitting Diodes (QLED's), Light emitting electrochemical cells.

Self-learning: *Properties and functions of Silicon (Si), Germanium (Ge), Copper (Cu), Aluminium (Al), and Brominated flame retardants in computers.*

MEMORY DEVICES

Memory device: A memory device is a piece of hardware used to store data. Most electronic devices such as computers, mobile phones, tablets, etc all have a storage device that stores data and /or programs.

Basic Concepts of Electronic Memory:

An electronic memory device is a form of semiconductor storage which is fast in response and compact in size and can be read and written when coupled with a central processing unit (CPU, a processor). In conventional silicon-based electronic memory, data are stored based on the amount of charge stored in the memory cells. Organic/polymer electronic memory stores data in an entirely different way, for instance, based on different electrical conductivity states (ON and OFF states) in response to an applied electric field. Organic/polymer electronic memory is likely to be an alternative or at least a supplementary technology to conventional semiconductor electronic memory.

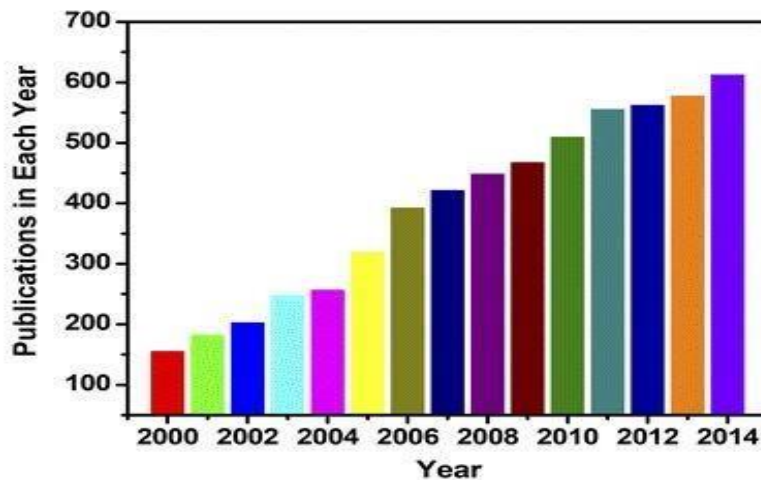
History of Organic/Polymer Electronic Memory Devices

Different forms of storage, based on various natural phenomena, have been reported since the 1940s. In the 1960s, there was a great interest in the electrical properties of amorphous semiconductors and disordered structures, arising from their

unusual electrical properties which also make them promising materials for device applications.

1. During 1968-70, bistable electrical conductivity and resistivity phenomenon was observed in Pb/polydivinyl benzene, tetracene, phthalocyanines and polystyrene polymer materials. All these materials showed bistable switching and memory switching effects. But the performance was not satisfactory for practical applications.
2. In 1980s, two stable ferroelectric polarization states were discovered in polymers. Thin films of ferroelectric polymer materials can be repeatedly switched between two stable and are capable of exhibiting non-volatile memory effects. But they required very high operating voltage of 30 V.
3. In 1995, ferroelectric polymer films as thin as 1 nm were fabricated. These required just 1 V to switch between two states. Since, then Polymer ferroelectric random-access memory (FeRAM) was developed as a promising memory technology.
4. In 2001, an organic field-effect transistors (OFETs) memory device was demonstrated using a sexithiophene oligomer as the conductor and a ferroelectric organic polymer material as gate insulator.
5. In 2003, a WORM type memory device was developed consisting of a thin film p-i-n silicon diode and a mixture of two conductive polymers, poly(ethylene dioxythiophene and poly(styrene sulfonic acid).
6. In 2004, ultrathin film organic materials were discovered with multilevel conductivity states. Poly[2-methoxy-5-(2-ethyl-hexyloxy)-1,4-phenylene vinylene polymer showed one low- and three high-conducting states. All four accessible states have associated memory effects for data-storage applications. To achieve ultrahigh density memory devices, organic materials with multilevel stable states are highly desirable.
7. Devices demonstrating multistability where more than two conducting states can be programmed into a single switching element will dramatically increase the amount of data stored per area or volume. Further progress in the development of multilevel organic polymer memory has been made in recent years.

The International Technology Roadmap for Semiconductors (ITRS) has identified polymer memory as an emerging memory technology since the year 2005.



CLASSIFICATION OF ELECTRONIC MEMORY DEVICES

According to the device structure, electronic memory devices can be divided into four primary categories: transistor-type, capacitor-type, resistor-type and charge transfer-type.

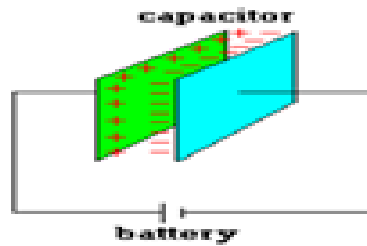
1. Transistor-type

Transistors are made from silicon, a semiconductor. It is converted to p-type and n-type semiconductor by doping trivalent and pentavalent impurities. Transistors are made using p-type and n-type semiconductor. A transistor is a miniature electronic component that can work either as an amplifier or a switch. A computer memory chip consists of billions of transistors, each transistor is working as a switch, which can be switched ON or OFF. Each transistor can be in two different states and store two different numbers, ZERO and ONE. Since chip is made of billions of such transistors and can store billions of Zeros and Ones, and almost every number and letter can be stored.

2. CAPACITOR- TYPE ELECTRONIC MEMORY DEVICE

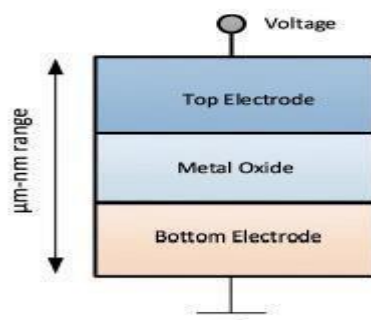
Capacitors have two parallel plate electrodes and charges are stored in these electrodes under an applied electric field. Data can be stored in these devices based on different charge stored in the cell. Charges stored in the cell maintain electric polarization that can be switched between two stable states by an external electric

field. Organic and polymeric ferroelectric materials can be used in capacitor-type electronic memory device.



3. RESISTOR-TYPE ELECTRONIC MEMORY DEVICE

Resistor-type electronic memory usually has a simple structure, having a metal-insulator-metal structure generally referred to as MIM structure. The structure comprises of an insulating layer (I) sandwiched between the two metal(M) electrodes and supported on a substrate (glass, silicon wafer, plastic, or metal foil). Initially, the device is under high resistance state or “OFF” and logically “0” state, when resistance changed or under external applied field changes to low resistance state or “ON” logical value “1”.



4. CHARGE TRANSFER TYPE ELECTRONIC MEMORY DEVICE

This type of electronic device is based on the Charge Transfer Effects of a charge transfer complex. A charge transfer (CT) complex consists of two parts, one electron donor and other an electron acceptor. It is also called as a donor-acceptor (D-A) complex. The conductivity of a CT complex is dependent on the ionic binding between the D-A components. In CT complex, a partial transfer of charges occurs from donor part to the acceptor part. This results in difference in conductivity. CT complexes exhibit bistable states due to difference in conductivity. This behavior is used to design molecular electronic devices.

ORGANIC MEMORY DEVICES

Organic electronic memory device stores data based on different electrical conductivity states (ON and OFF states) in response to an applied electric field.

TYPES OF ORGANIC MEMORY MATERIALS

There are three classes of materials which can exhibit bistable states and are used in organic memory devices. They are:

- 1) Organic molecules,
- 2) Polymeric materials,
- 3) Organic- inorganic hybrid materials.

Under each category, lot of different types of molecules exhibiting memory effect are available. Few of them are described here:

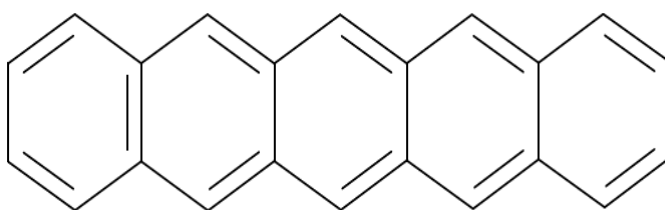
1) Organic molecules -Small organic molecules containing both an electron donor and an electron acceptor are an important type of material for organic electronic memory devices.

a) The p Type Organic Semiconductor Material “Pentacene”

An Organic molecule with π conjugated system and possesses holes as major charge carrier is called p-type semiconductor.

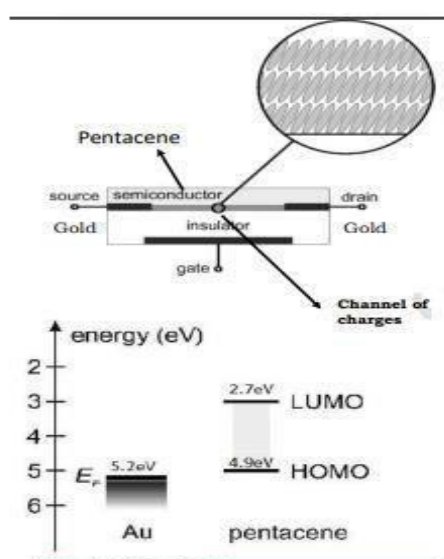
Example: Pentacene

Pentacene is a linear aromatic hydrocarbon formed by the fusion of five benzene rings. The extended π -system allows the continuous delocalization of π -electrons and there is a lateral over lapping of pi-electrons between the molecules.



Pentacene

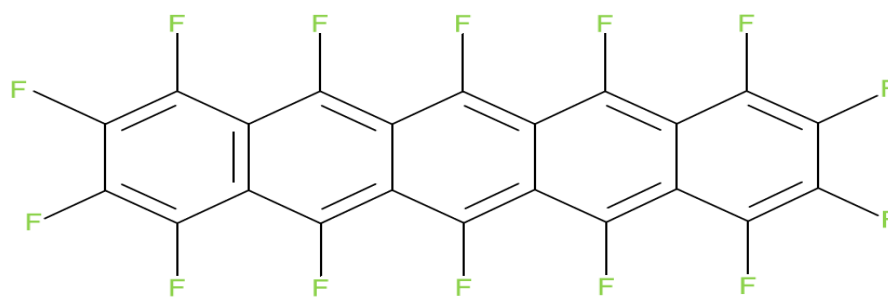
To show that Pentacene is a p-Type semiconductor.



Consider OTFT (Optical Thin Film Transistor) as shown above. Source and drain were made of gold and semiconductor was Pentacene. When a Positive voltage is applied to the gate negative charges are induced at the source electrode (Au). Since, Fermi level of gold is a way from LUMO of Pentacene, electron flow cannot take place. When a negative voltage is applied to the gate holes are injected from source to semiconductor because Fermi level energy of gold is close to energy of HOMO of Pentacene. A conducting channel is formed at the insulator and semiconductor interface and allows the movement of charge carriers holes from source to drain when secondary voltage is applied. Therefore, Pentacene is considered as p-type semiconductor.

b) The n-type organic semiconducting material Perfluoropentacene

An Organic molecule with π conjugated system with electron withdrawing substituent groups and possess electrons as major charge carrier is called n-type semiconductor. When a p-type semiconductor Pentacene is modified with electronegative fluorine atoms to lower the Lowest Unoccupied Molecular Orbital (LUMO) energy levels of materials for electron injection and transport. The HOMO-LUMO gaps of perfluorinated pentacene are smaller than those of the corresponding Pentacene.



Perfluoropentacene

To show that Perfluoropentacene is a n-Type semiconductor.

Consider OTFT (Optical Thin Film Transistor) as shown above. Source and drain were made of gold and semiconductor was Perfluoropentacene. The HOMO-LUMO gaps are 2.07eV for pentacene and 1.95eV for perfluoropentacene. When a Positive voltage is applied to the gate negative charges are induced at the source electrode (Au). The source and drain electrodes inject electrons in its LUMO level. This charge forms a conducting channel at the insulator and semiconductor interface and allows the charge carriers electron from source to drain. Therefore, Perfluoropentacene is considered as n-type semiconductor.

2) Polymer molecules

Polymeric material for Organic memory device

A volatile memory is a type of memory which cannot sustain the two distinct electronic states without an external electronic power supply, that is, the written data will disappear. A non-volatile memory can sustain the two distinct states without the power supply, that is, the written data will not disappear when an external electronic power supply is removed. Understanding the relationship between the chemical structure and memory properties is a subject of utmost importance in the development of polymer memory materials. One such polymer used for organic memory device is Polyimide with Donor-Triphenyl amine and Acceptor-phthalimide.

Donor: Triphenyl Amine group (TPA) Acceptor: Phthalimide group

Hexa fluoro isopropylidene (6F): Increases the solubility of PI

The donors and acceptors of PIs contribute to the electronic transition based on an induced charge transfer (CT) effect under an applied electric field. When an electric

field more than threshold energy is applied, the electrons of the HOMO (TPA unit) is excited to LUMO. The energy of LUMO of donor and acceptor are similar and therefore, after excitation the electron transferred to LUMO (acceptor), generating a CT state. This permits the generation of holes in the HOMO, which produces the open channel for the charge carriers to migrate through. Therefore, Field-induced charge transfer from triphenyl amine to phthalimide exhibit the switching behavior (bistable states ON/OFF). This device exhibits dynamic random-access memory (DRAM) behavior with an ON/OFF current ratio of upto 10^5 .

3) Organic-inorganic hybrid materials

Generally, organic-inorganic hybrid materials are composed of organic layers containing inorganic materials. Inorganic materials used are allotropes of carbon like fullerenes, carbon nanotubes, graphene and metal nanoparticles, semiconductor nanoparticles and inorganic quantum dots (QDs).

1) Organic-Carbon Allotrope Hybrid Materials

Polymers containing electron donors, such as thiophene, fluorene, carbazole and aniline derivatives can be combined with Fullerenes to obtain a charge transfer hybrid material with donor-acceptor ability and electrical bistable states. Fullerenes exhibit high electron-withdrawing ability and can capture up to six electrons. They are used in WORM memory effect devices.

2) Organic-Inorganic Nanocomposites

These are the hybrid electronic memory devices in which organic polymer with appropriate functional group is clubbed with metal nanoparticles, quantum dots and metal oxide nanoparticles. An example is a composite of 8-hydroxyquinoline-containing polymer with gold nanoparticle sandwiched between two metal electrodes. Bistable electronic transition states are observed when an electric field is applied due to charge transfer between the Au nanoparticles and 8-hydroxyquinoline.

DISPLAY SYSTEMS

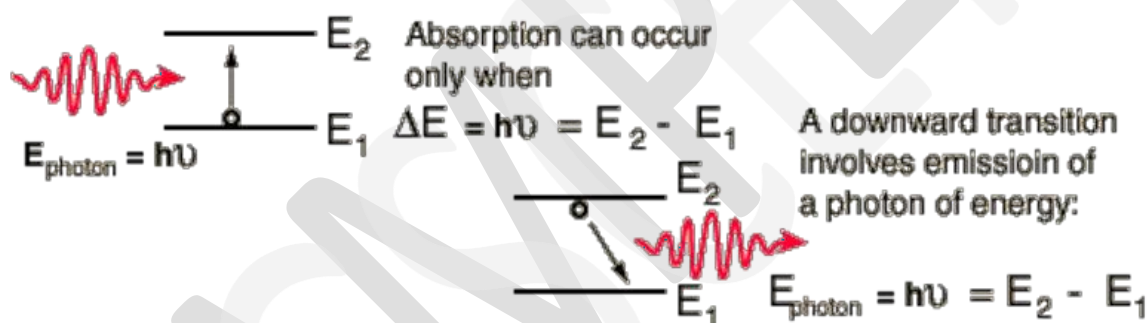
Display: Display is an output device used to present visual information.

Optoelectronics

Optoelectronics is the communication between optics and electronics, which includes the study, design and manufacture of a hardware device that converts electrical energy into light and light into energy through semiconductors.

Optoelectronic devices: A hardware device that converts electrical energy into light and light into energy through semiconductors.

Optoelectronic devices are primarily transducers i.e., they can convert one energy form to another. They can also detect light and transform light signals to electrical signals for processing by a computer.



Optoelectronic devices are special types of semiconductor devices that are able to convert light energy to electrical energy or electrical energy to light energy. If the photon has an energy larger than the energy a gap, the photon will be absorbed by the semiconductor, exciting an electron from the valence band into the conduction band, where it is free to move. A free hole is left behind in the valence band. When the excited electron is returning to valence band, extra photon energy is emitted in the form a light. This principle is used in Optoelectronic devices.

Photoactive and electro active organic materials:

Organic semiconductors used in electronic and optoelectronic devices are called as electroactive and Photoactive materials. Photoactive and electroactive organic materials are the semiconductors composed of π -electron systems.

Working Principle

Photoactive and electroactive material absorb and emit light in the UV to IR region.

Display system (OLED) consisting of photoactive and electroactive material absorb light and allows an electron to jump from HOMO of a Donor to LUMO of an Acceptor. This phenomenon generates and transport charge carriers.

In an OLED device, the light-emitting layer is excited by the recombination energy of electrons from the cathode and holes from the anode, and then the light-emitting layer emits light when returning to the ground state. One of the electrodes consists of transparent material in order to extract light from the light emitting layer.

NANOMATERIALS USED IN OPTOELECTRONIC DEVICES

Nanomaterials: Nanomaterials are those materials whose characteristic length lies within nanometer range (1-100nm). The properties of nanomaterials are different from bulk materials due to:

1. Quantum Confinement effect
2. Increased surface area to volume ratio

The improved electronic properties yielded for nanostructured silicon in comparison to its bulk, which led the use of Silicon Nanocrystals in electronics and optoelectronics fields.

Special properties of Silicon Nano crystals for optoelectronics

1. Silicon Nanocrystal has wider bandgap energy due to quantum confinement.
2. Si NCs shows higher light emission property (Photoluminescence)
3. SiNCs exhibit quantum yield of more than 60%.
4. Si-NCs exhibit tunable electronic structure
5. Larger surface area-volume ratio.

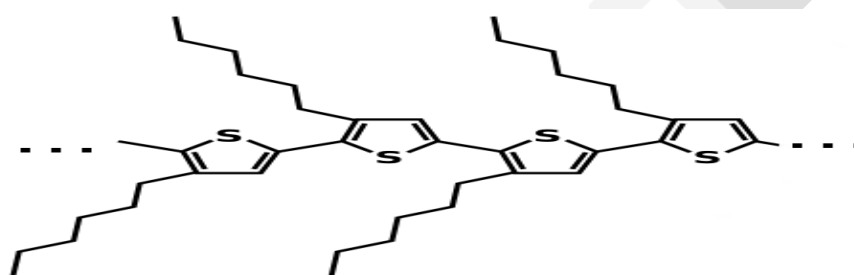
Applications:

1. Si NCs are used in neuro morphic computing and down-shifting in photovoltaics
2. Si NCs are used in the construction of novel solar cells, photodetectors and optoelectronic synaptic devices.

Organic materials for Optoelectronic devices [Light absorbing materials – Polythiophenes] (P3HT)

Polythiophenes are an important class of conjugated polymers, environmentally and thermally stable material. Chemical structure of P3HT Poly(3-hexylthiophene) is a polymer with chemical formula $(C_{10}H_{14}S)_n$. It is a polythiophene with a short alkyl group on each repeat unit.

Highly ordered (P3HT) are composed of closely packed, p-p stacked (p-p distance of 0.33nm) fully extended chains which are oriented perpendicular to the substrate.



structure of P3HT Poly(3-hexylthiophene)

Properties:

1. P3HT is a semiconducting polymer with high stability and exhibits conductivity due to holes therefore considered as p-type semiconductor.
2. Poly-3-hexylthiophene(P3HT) have great capability as light-absorbing materials in organic electronic devices.
3. P3HT has a crystalline structure and good charge-transport properties required for Optoelectronics.
4. P3HT has a direct-allowed optical transition with a fundamental energy gap of 2.14eV.
5. Fundamental bandgap of P3HT is 490nm visible region, corresponding to $\pi \rightarrow \pi^*$ transition, giving electron-hole pair.
6. P3HT indicate that an increase in the conductivity is associated with an increase in the degree of Crystallinity.

Applications:

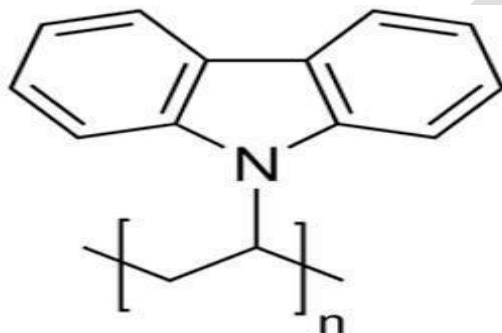
- P3HT-ITO forms a p-n junction permit the charge carriers to move in opposite direction and hence, used in Photovoltaic devices.
- It can be used as a positive electrode in Lithium batteries.
- Used in the construction of Organic Solar Cells.

MATERIALS FOR MEMORY AND DISPLAY SYSTEMS

- Manufacture of smart windows.
- Used in the fabrication new types of memory devices.

Light emitting material-Poly [9-vinyl carbazole] (PVK)]

Poly (N-vinyl carbazole) (PVK) is one of the highly process able polymers as hole conducting material and therefore used as an efficient hole transport material to prepare highly efficient and stable planar hetero junction perovskite solar cells.



Structure of poly(9-vinylcarbazole) (PVK) Applications

- PVK has been commonly used in OLEDs, light harvesting applications, photo refractive polymer composites and memory devices.
- Used in the fabrication of light-emitting diodes and laser printers.
- Used in the fabrication of organic solarcells when combined with TiO on glass substrate.
- Used in the fabrication of solarcells when combined with Perovskite materials.
- PVK-Perovskite junction is used in Light-Emitting Diodes with Enhanced Efficiency and Stability.

LIQUID CRYSTAL DISPLAYS

The **liquid crystals** are a unique state of matter between solid(crystalline)and liquid(isotropic)phases.

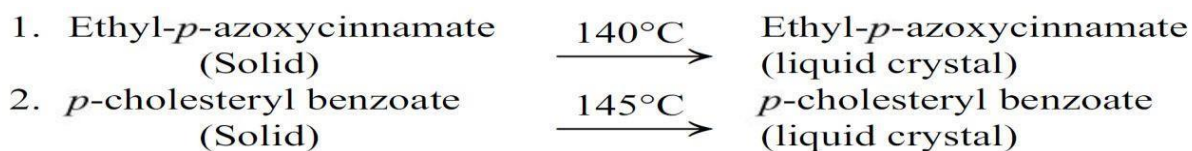
Classification of Liquid crystal

Liquid crystals are classified into two main categories, namely,

1. Thermotropic Liquid Crystals
2. Lyotropic Liquid Crystals

1) Thermotropic Liquid Crystals:

A liquid crystal is said to be thermotropic if molecules orientation is dependent on the temperature. By increasing the temperature, the increase in energy and there by movement of constituent molecules, will induce phase changes.



Thermotropic liquid crystals may be further classified as:

- Nematic liquid crystals
- Chiral Nematic liquid crystals
- Smectic liquid crystals
- Discotic Liquid Crystalline Phase

1. Nematic liquid crystals (or thread-like liquid crystals)

The molecules move either sideways or up and down. Since the molecules are oriented in one direction, they exhibit anisotropy. In this case, the molecules are readily aligned in the same direction in the presence of electric and magnetic fields. The alignment of molecules is temperature sensitive – as the temperature is increased, the degree of orientation of the nematic crystals decreases and they change into isotropic liquids.

Examples: *p*-azoxy phenetole, anisaldazine.

2. Chiral (Twisted) Nematic liquid crystals

In chiral nematic or twisted nematic liquid crystals (TNLC) all the molecules are roughly parallel to one another, the molecules in the chiral nematic phase arrange themselves to form a helical configuration.

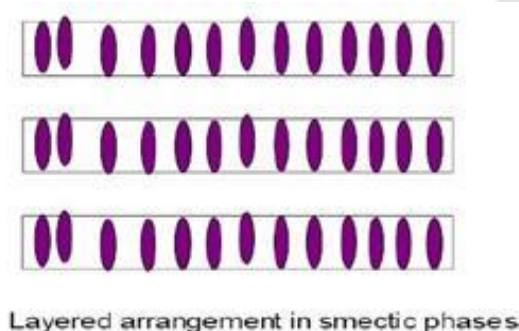
Examples: cholesteryl formate, cholesteryl myristate, and cholesteryl benzoate.



3. Smectic liquid crystals

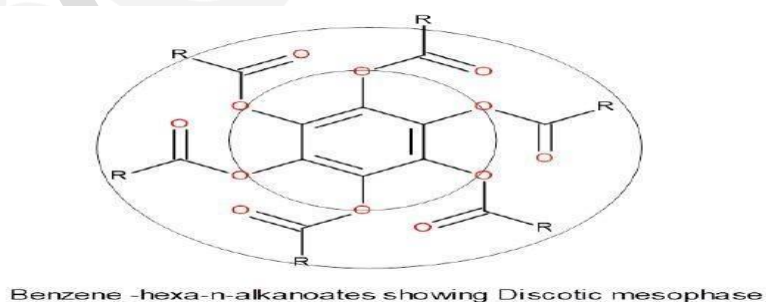
Substances that form smectic phases are soap-like (in Greek, smectos means soap). In smectic liquid crystals, there is a small amount of orientational order and also a small amount of positional order. The molecules are arranged in regularly spaced layers (positional order). Within the layer they tend to point along the director (orientational order).

Examples: 4-cyanobenzylidene-4-n-octyloxyaniline.



4. Discotic Liquid Crystalline Phase

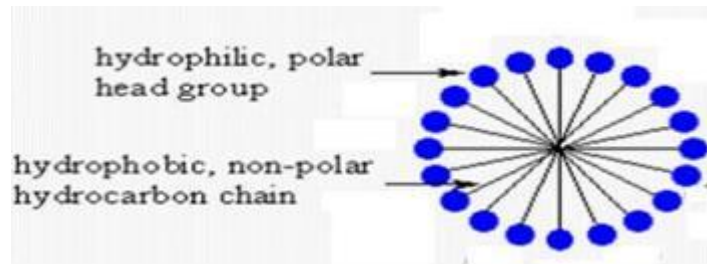
Molecules with a disk- or plate-like structure demonstrate the liquid crystal phase. These are known as columnar or discotic liquid crystals. In the centre region, these molecules have stiff structures like benzene and triphenylene. Because there is orientational order but no positional order in the simplest discotic phase, it is often referred to as the discotic nematic phase.



2. Lyotropic Liquid Crystals

The orientational behavior of lyotropic crystals is a function of concentration and solvent. These molecules are amphiphilic—they have both hydrophilic and hydrophobic ends in their molecules. The hydrophilic end is attracted towards

water, whereas the hydrophobic end is water repellent and attracted towards non-polar solvents. At low concentrations, these molecules are randomly oriented but as the concentration increases, the molecules start arranging themselves. Cell membranes and cell walls are examples of lyotropic liquid crystals. Soaps and detergents form lyotropic crystals when they combine with water.



Properties of liquid crystals

1. They exhibit optical anisotropy which is defined as the difference between refractive index parallel to the director and refractive index perpendicular to the director. These two properties are important for the electro-optic effects in liquid crystals.
2. The intermolecular forces are rather weak and can be perturbed by an applied electric field.
3. Because the molecules are polar, they interact with an electric field, which causes them to change their orientation slightly.
4. Liquid Crystal can flow like a liquid, due to loss of positional order.
5. Liquid crystal is optically birefringent, due to its orientation order.
6. Liquid crystals, like all other kinds of matter, exhibits thermal expansion (Thermal Imaging).

APPLICATIONS OF LIQUID CRYSTALS IN DISPLAY SYSTEMS

Liquid crystal displays operate at low voltages (a few volts) and consume less power as compared to other display and hence are used in various fields, including electronics, optics, displays, sensors, and medicine. Some of the major applications of liquid crystals are:

1. Watches, calculators, mobile telephones, laptop computers and related electronic gadgets.
2. Indicators in automobile dashboards, airplane cockpits, traffic signals, advertisement boards and petrol pump indicators.

3. Blood pressure instruments, digital thermometers, and TV channel indicators.
4. pH meters, conductometers, colorimeters, potentiometers, and other analytical instruments.

ORGANIC LIGHT EMITTING DIODES (OLED'S)

“OLEDs are thin film devices consisting of a stack of organic layers sandwiched between two electrodes. OLEDs operate by converting electrical current into light via an organic emitter”.

Properties of Organic Light Emitting Diodes (OLED's)

1. OLED devices have solid and planar structure. Therefore, OLED display panels are very thin, flat and lightweight.
2. OLED devices have self-emission property and hence their devices have high contrast ratios and wide viewing angles, which are very significant factors for displays.
3. The response time of OLEDs is as fast as micro- or nanosecond order. Therefore, OLED displays can produce sharp moving images.
4. OLEDs have a wide viewing angle, which means that the image quality is maintained even when viewed from different angles.
5. OLEDs are energy efficient, as they do not require a back light like traditional LCD displays, resulting in lower power consumption.
6. OLEDs are self-emissive, which means that they do not require a separate light source, resulting in a thinner display.
7. OLEDs have a long lifespan, as they do not contain a back light that can degrade over time, resulting in a longer-lasting display.

Application of Organic Light Emitting Diodes (OLED's)

1. OLED displays are used in televisions, monitors, smartphones, and other electronic devices.
2. OLEDs can also be used as a source of lighting in various applications, including automotive lighting, street lighting, and architectural lighting.
3. The thin and flexible nature of OLEDs makes them suitable for use in wearable devices, such as smart watches and fitness trackers.
4. OLEDs can be used in automotive applications, such as dashboard displays, and interior lighting.
5. OLEDs can be used in medical applications, such as in surgical lighting and medical imaging.

6. They offer bright and highly accurate lighting options that can help improve medical procedures and diagnosis.

QUANTUM LIGHT EMITTING DIODES (QD-LEDS)

Quantum dot light emitting diodes are a form of light emitting technology and consist of nano- scale crystals that can provide an alternative for applications such as display technology". The structure of QD-LED is like the fundamental design of OLED. But the difference is that the light emitting is from the QDs, such as cadmium selenide (CdSe) nanocrystals.

Properties of QLED

1. QD-LEDs are capable of producing highly accurate and vibrant colors due to their use of quantum dots, which emit light of a specific color when they are excited by a light source or an electrical current.
2. QD-LEDs are more energy-efficient than traditional LCD displays because they do not require as much back lighting.
3. QD-LED displays have high contrast ratios, which means that the difference between the darkest and brightest areas of the display is greater, resulting in more detailed and life like images.
4. QD-LEDs have a longer lifespan than traditional LCD displays because they do not suffer from the same issues of back light burn out or color fading over time.
5. QD-LEDs displays have fast response times, which mean that they can display fast-moving images without motion blur or ghosting.
6. QD-LEDs can be made on flexible substrates, which allows for the creation of flexible displays that can be bent or curved.

Applications of QD-LED

- 1) QD-LED displays are commonly used in televisions, monitors, smartphones, and other electronic devices. They offer superior image quality and color accuracy compared to traditional LCD displays.
- 2) QD-LEDs can also be used as a source of lighting invarious applications, including automotive lighting, street lighting, and architectural lighting. They offer energy-efficient and highly customizable lighting options.

- 3) QD-LEDs can be used in medical imaging applications, such as in MRI machines, to produce high-resolution and accurate images.
- 4) QD-LED displays are suitable for use in virtual and augmented reality applications due to their ability to produce vibrant and accurate colors, which can enhance the immersive experience.
- 5) QD-LED displays can be used in advertising displays, such as digital billboards and signage, to produce high-quality and eye-catching visuals.

Light-emitting electrochemical cell

Light-emitting electrochemical cells (LEC) are solid-state lighting devices that convert electric current to light within electroluminescent organic semiconductors.

- LECs are best known for their simple architecture and easy, low-cost fabrication techniques.
- The key feature of their fabrication is the use of air stable electrodes and a single active layer consisting of mobile ions that enable efficient charge injection and transport processes within LEC devices.
- LEC devices can be operated at low voltages with high efficiencies, contributing to their widespread interest.

Applications

- Electroluminescent inks and stickers
- Used as light emitting materials.